

Average Speed Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links distance (d), speed (s), and time (t). Cover the unknown to read the formula: $d = s \times t$, $s = d \div t$, and $t = d \div s$.

Round 1 — Retrieve It

1. What three quantities does the speed triangle connect?

2. Write the formula for distance.

3. Write the formula for speed.

4. Which variable do you cover to find time, and what formula results?

Round 2 — Sketch It

Draw the speed triangle. Put distance on top; put speed and time on the bottom. Circle the two quantities you would multiply to find distance.

Average Speed — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A car travels at 20 m/s for 5 s. How far does it go?

2. A runner covers 150 m in 30 s. What is the average speed?

3. A cyclist rides 120 m at 8 m/s. How long does it take?

Round 4 — Reason from Evidence

Two buses each cover the same 300 m. Bus A takes 20 s; Bus B takes 25 s. Use the triangle to find each bus's speed, then say which bus is faster and how you know.

Average Speed — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain how the speed triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C

Connect

Connect speed to a distance–time graph. What does a steeper line mean, and how does that relate to $s = d \div t$?

E

Extend

Create your own problem that asks for time. Give the numbers and show the work using $t = d \div s$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Average Speed

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Distance, speed, and time.
2. $d = s \times t$.
3. $s = d \div t$.
4. Cover t ; $t = d \div s$.

Sketch It — look for

Distance on top, with speed and time on the bottom. Distance is found by multiplying speed $\times$ time (the two bottom quantities).

Apply It

1. $d = s \times t = 20 \times 5 = \mathbf{100 \text{ m}}$.
2. $s = d \div t = 150 \div 30 = \mathbf{5 \text{ m/s}}$.
3. $t = d \div s = 120 \div 8 = \mathbf{15 \text{ s}}$.

Reason from Evidence — look for

Bus A: $s = 300 \div 20 = 15 \text{ m/s}$. Bus B: $s = 300 \div 25 = 12 \text{ m/s}$. Bus A is faster because it covers the same distance in less time.

ACE — Extend look-for

Many answers possible. Check that the problem asks for time and the work correctly uses $t = d \div s$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.